

Human Reproduction

Year 9 Science · exam sprint · 2 hours

- 0:04 Loop 1 · Sex cells and the two systems
- 0:26 Loop 2 · Hormones and the menstrual cycle
- 0:49 Loop 3 · Fertilisation, chromosomes, twins
- 1:12 Loop 4 · Pregnancy, birth, fertility
- 1:44 Mini mock

Teaching line, not a question slide.
Plan: opening 0:00, Loop 1 0:04, Loop 2 0:26, Loop 3 0:49, Loop 4 1:12, lose marks and summary 1:42, mini mock 1:44, homework 1:59.
Each loop ends with about 6 minutes of writing on the In-class work sheet (Parts A to D). Part E is the mini mock.
Say it:

reproduction = ree-pruh-DUK-shun
hormones = HOR-mohnz
menstrual = MEN-stroo-ul
fertilisation = fer-tih-ly-ZAY-shun
chromosomes = KROH-muh-sohmz
pregnancy = PREG-nun-see
fertility = fer-TIL-ih-tee

Do Now

- 1 Name the male sex cell and the female sex cell.
- 2 Which organ makes the female sex cells?
- 3 What is the joining of the two sex cells called?
- 4 How many chromosomes are in a human body cell?
- 5 Which organ passes oxygen from the mother to the foetus?

1. sperm (male); egg or ovum (female)
 2. the ovaries
 3. fertilisation
 4. 46 (23 pairs)
 5. the placenta
- Diagnostic:

1 to 3 wrong: go slowly through Loop 1. 4 wrong: spend longer on the chromosome slide in Loop 3. 5 wrong: spend longer on the placenta slides in Loop 4.

Say it:

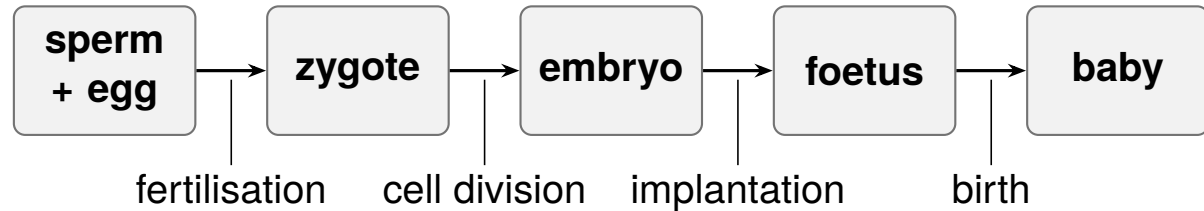
ovum = OH-vum

ovaries = OH-vuh-reez

placenta = pluh-SEN-tuh

foetus = FEE-tus

The one idea: from two cells to a baby



Every question is about one of these steps.

For each step, know the cell, the place and the process.

Which step happens in the oviduct?

Teaching line, not a question slide.

Q fertilisation happens in the oviduct; the zygote starts to divide there on its way to the uterus

Say it:

zygote = ZY-goht

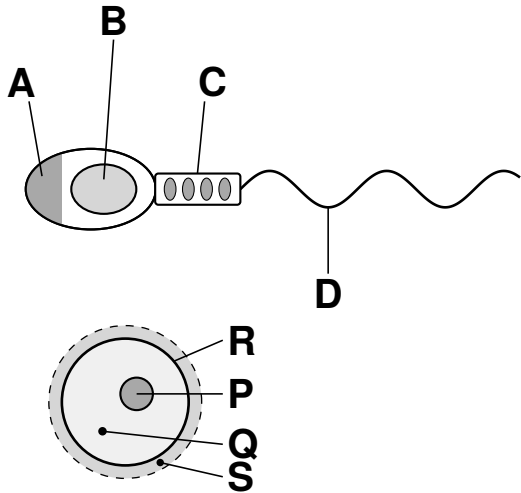
embryo = EM-bree-oh

implantation = im-plan-TAY-shun

oviduct = OH-vih-dukt

uterus = YOO-tuh-rus

Quick review: Sex cells



- 1 Name parts A to D and parts P to S.
- 2 How many chromosomes are in each nucleus?
- 3 Which cell is larger?
Which cell can move?

1. A tip of the head (it holds enzymes), B nucleus, C middle piece (it holds mitochondria), D tail; P nucleus, Q cytoplasm, R cell membrane, S jelly layer
2. 23 in each
3. the egg is much larger; only the sperm can move: it swims with its tail

Say it:

nucleus = NYOO-klee-us

cytoplasm = SY-toh-plaz-um

membrane = MEM-brayn

enzymes = EN-zymz

mitochondria = my-toh-KON-dree-uh

Adaptations: the feature and how it helps

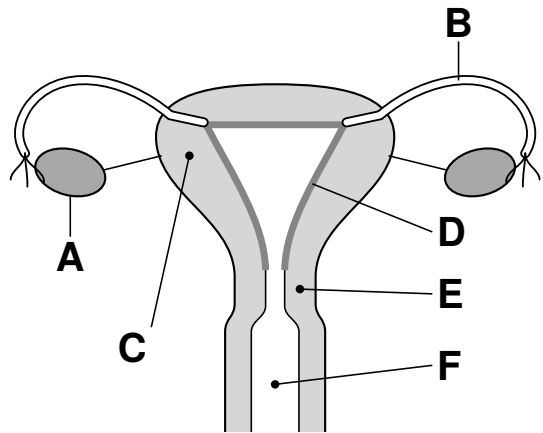
Cell	Feature	How it helps
sperm	long tail	
sperm	many mitochondria	
sperm	enzymes in the head	
egg	large cytoplasm	
egg	membrane changes	

Why does a man make millions of sperm, but a woman releases one egg a month?

Teaching line, not a question slide.
Gaps: it swims to the egg; they release energy for swimming; they break through the jelly layer and membrane of the egg; it is a food store for the first cell divisions; it stops any other sperm getting in.

Q very few sperm survive the journey to the oviduct, so millions give a better chance that one meets the egg; one egg is enough because it stays inside the body
Say it:
adaptations = ad-ap-TAY-shunz

Quick review: Female system, front view



- 1 Name parts A to F.
- 2 Where are eggs made?
- 3 Where does a foetus develop?
- 4 Which part is a ring of muscle?

1. A ovary, B oviduct (fallopian tube), C uterus, D lining of the uterus (endometrium), E cervix, F vagina
2. in the ovaries
3. in the uterus
4. the cervix

Say it:

ovary = OH-vuh-ree

fallopian = fuh-LOH-pee-un

endometrium = en-doh-MEE-tree-um

cervix = SUR-viks

vagina = vuh-JY-nuh

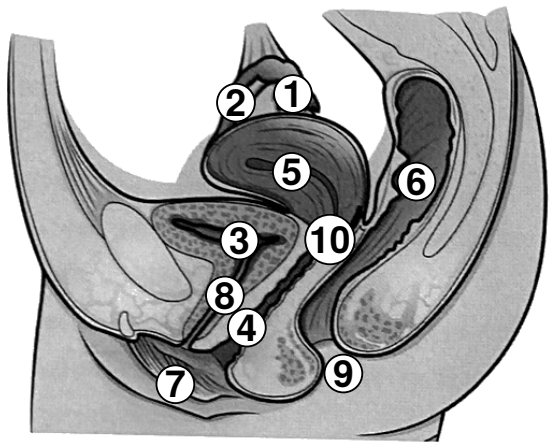
Female system: the job of each part

Part	Job
ovary	makes _____ and two hormones
oviduct	carries the egg; _____ happens here
uterus	the _____ develops here
lining	gets thick so that an _____ can implant
cervix	a ring of _____ at the base of the uterus
vagina	receives _____; the baby is born through it

Teaching line, not a question slide.
Gaps: eggs; fertilisation; foetus; embryo; muscle; sperm (semen).
The two hormones from the ovary are oestrogen and progesterone.

Say it:
oestrogen = EE-struh-jun
progesterone = proh-JES-tuh-rohn
semen = SEE-mun

Quick review: Female system, side view



anus · bladder · cervix ·
fallopian tube · ovary · rectum ·
urethra · uterus · vagina · vulva

- 1 Match the words to parts 1 to 10.
- 2 Which four parts are not reproductive organs?

1. 1 ovary, 2 fallopian tube, 3 bladder, 4 vagina, 5 uterus, 6 rectum, 7 vulva, 8 urethra, 9 anus, 10 cervix

2. bladder and urethra (they store and carry urine); rectum and anus (digestive system)

Say it:

anus = AY-nus

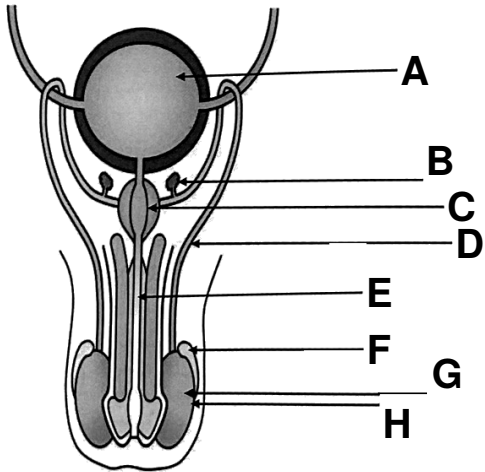
rectum = REK-tum

urethra = yoo-REE-thruh

vulva = VUL-vuh

urine = YOOR-in

Quick review: Male system



- 1 Name parts A to H.
- 2 Where are sperm made?
- 3 Which tube carries both urine and semen?

1. A bladder, B seminal vesicle, C prostate gland, D sperm duct (vas deferens), E urethra, F epididymis, G testis, H scrotum

2. in the testes

3. the urethra

Say it:

seminal = SEM-ih-nul

vesicle = VES-ih-kul

prostate = PROS-tayt

vas deferens = vas-DEF-uh-renz

epididymis = ep-ih-DID-ih-mis

testis = TES-tis

testes = TES-teez

scrotum = SKROH-tum

The path of a sperm cell

Teaching line, not a question slide.

Order: B, D, C, E, A. Gap: semen.

Q sperm are made best at a temperature a little below body

temperature; in the scrotum the testes stay cooler

Say it:

penis = PEE-nis

Put the cards in order.

A The urethra carries semen out through the penis.

B Sperm are made in a testis.

C Sperm travel along the sperm duct.

D Sperm are stored in the epididymis.

E Glands add fluid: sperm + fluid = _____.

Why are the testes held outside the body?

Male system: the job of each part

Teaching line, not a question slide.
Gaps: cooler; sperm, testosterone; stores; urethra; fluid; vagina.

The glands are the seminal vesicles and the prostate gland.
Say it:
testosterone = tes-TOS-tuh-rohn

Part	Job
scrotum	holds the testes outside the body, so they stay _____
testes	make _____ and the hormone _____
epididymis	_____ the sperm
sperm duct	carries sperm to the _____
glands	add _____ that feeds the sperm
penis	puts semen into the _____

Spot the error (1)

Describe where sex cells are made and where a foetus develops.

Sam's answer

Eggs are made in the uterus and travel down the oviduct. The cervix is the place where the foetus develops. In a man, sperm are made in the testes and stored in the scrotum.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. eggs are made in the ovaries, not in the uterus (L4)
2. the foetus develops in the uterus; the cervix is the ring of muscle at its base (L4)
3. sperm are stored in the epididymis; the scrotum is the bag of skin

that holds the testes (L4)

Short solution:

Eggs are made in the ovaries and travel down the oviduct. The uterus is the place where the foetus develops. In a man, sperm are made in the testes and stored in the epididymis.

Practice

Q1 State two differences between a sperm cell and an egg cell. **[A]**

Q2 Explain why the testes are held outside the body in the scrotum. **[M]**

Q3 A woman's oviducts are both blocked. She still releases an egg each month. Explain why she cannot become pregnant naturally. **[E]**

Q1 any two: a sperm is much smaller; a sperm has a tail and can swim, an egg cannot move; an egg has a large food store in its cytoplasm; millions of sperm are made, one egg is released each month

Q2 sperm are made best at a temperature a little below body temperature; outside the body the testes stay cooler
Q3 fertilisation happens in the oviduct; the sperm cannot reach the egg and the egg cannot reach the uterus, so no zygote forms and nothing can implant

In-class work · Part A

- 1 Label the female reproductive system.
- 2 Complete the sentences about the male system.
- 3 Explain one adaptation of a sperm cell.

Write in full sentences. Use the science words.

Puberty: hormones switch it on

Teaching line, not a question slide.
Gaps: messengers; blood; testosterone; oestrogen; periods.
Q true

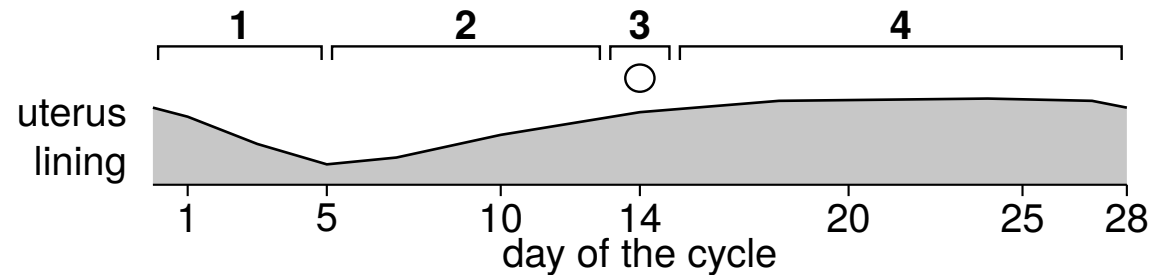
Both sexes: a growth spurt, and pubic and underarm hair.
Say it:
puberty = PYOO-buh-tee

Hormones are chemical _____, carried in the _____.

Boys		Girls	
hormone _____		_____	
made in	testes	ovaries	
changes	voice deepens	_____	start
also	sperm are made	breasts grow	

True or false? Girls usually start puberty before boys.

The menstrual cycle: stages 1 and 2



- 1** Days 1 to 5: the lining _____ and leaves through the vagina. This is a _____.
- 2** Days 5 to 13: the lining _____ again. An egg matures in an _____.

Teaching line, not a question slide.

Gaps: breaks down; period (menstruation); builds up; ovary.

Stage names: 1 menstrual, 2 follicular. In stage 2 a follicle in the ovary grows; FSH (follicle stimulating hormone) makes the egg inside it mature.

Say it:

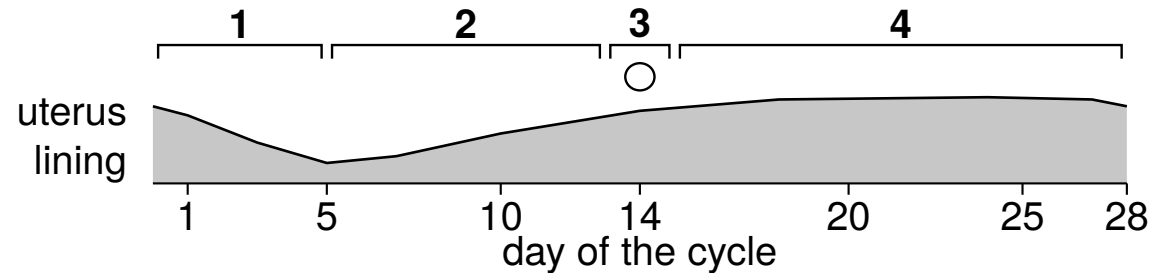
menstruation = men-stroo-AY-shun

follicular = fuh-LIK-yoo-luh

follicle = FOL-ih-kul

stimulating = STIM-yoo-lay-ting

The menstrual cycle: stages 3 and 4



3 About day 14: an egg is _____ from an ovary. This is _____.

4 Days 15 to 28: the lining stays _____, ready for a fertilised egg.

When is a woman most likely to become pregnant?

Teaching line, not a question slide.

Gaps: released; ovulation; thick.

Q around day 14: on the day of ovulation and the two days before and after it

Stage names: 3 ovulation, 4 luteal. With no fertilised egg the lining breaks down and the cycle starts again.

Say it:

ovulation = ov-yoo-LAY-shun

luteal = LOO-tee-ul

Four hormones, four jobs

Hormone	Made in	Job
FSH	pituitary gland	makes an egg _____ in the ovary
LH	_____	triggers _____ on day 14
oestrogen	ovary	_____ the lining of the uterus
progesterone	_____	keeps the lining _____

FSH = follicle stimulating hormone. LH = luteinising hormone.
Which two hormones peak at ovulation?

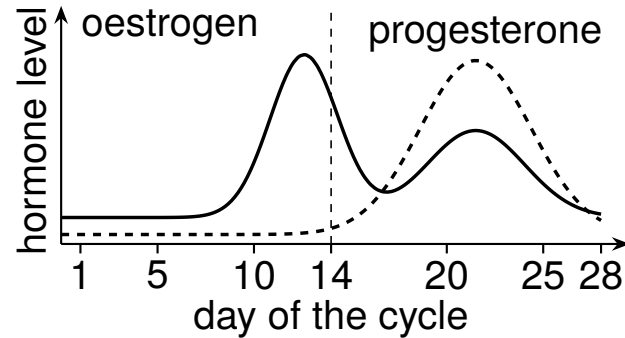
Teaching line, not a question slide.
Gaps: mature; pituitary gland, ovulation; builds up; ovary, thick.
Q FSH and LH

Say it:
pituitary = pih-TYOO-ih-tree
luteinising = LOO-tee-in-eye-zing

Read the graph: ovarian hormones

Teaching line, not a question slide.
1. oestrogen

2. progesterone
3. the lining breaks down and is lost: a period starts

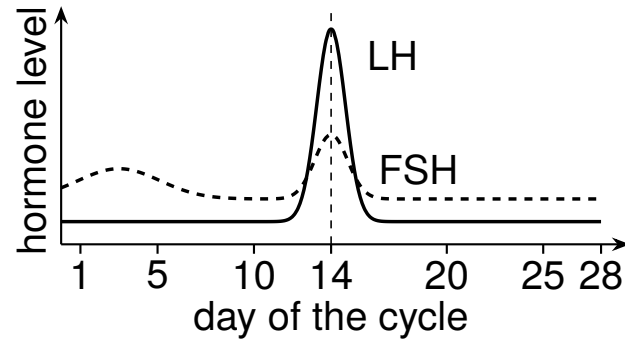


- 1 Which hormone peaks just before day 14?
- 2 Which hormone is highest at about day 21?
- 3 Both levels fall by day 28. What happens to the lining?

Read the graph: pituitary hormones

Teaching line, not a question slide.
1. day 14

2. ovulation: an egg is released from an ovary
3. it stays low and almost constant



- 1 On which day do LH and FSH peak?
- 2 What does the LH peak cause?
- 3 Describe the level of LH from day 1 to day 12.

No fertilised egg: write it in this order

Teaching line, not a question slide.

Gaps: implants; progesterone; breaks down; vagina.

Q the embryo implants, progesterone stays high and the lining stays thick, so there is no period

1 The egg is not fertilised, so no embryo _____ in the lining.

2 The level of _____ falls.

3 The lining of the uterus _____.

4 Blood and lining leave through the _____: a period.

A: lines 3–4 **M:** links the hormone to the lining **E:** the whole chain

What is different if the egg is fertilised?

Spot the error (2)

Describe ovulation and name one hormone that changes the lining of the uterus.

Sam's answer

Ovulation is when the lining of the uterus is lost. An egg is released on about day 28 of the cycle. Oestrogen is made in the pituitary gland and makes the lining thicker.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. ovulation is the release of an egg from an ovary; losing the lining is menstruation (L5)
2. the egg is released on about day 14, not day 28 (L5)
3. oestrogen is made in the ovaries, not in the pituitary gland (L6)

Short solution:

Ovulation is the release of an egg from an ovary, on about day 14 of the cycle. Oestrogen is made in the ovaries and makes the lining thicker.

Practice

Q1 A woman has a 28-day cycle. On which days is she most likely to become pregnant? **[A]**

Q2 Describe how the lining of the uterus changes between day 5 and day 14. Name the hormone that causes this. **[M]**

Q3 The level of progesterone stays high in a pregnant woman. Explain why this is important. **[E]**

Q1 about days 12 to 16: ovulation is on about day 14, and she is fertile for two days before and after it

Q2 the lining builds up and becomes thicker, with more blood

vessels; oestrogen

Q3 progesterone keeps the lining of the uterus thick, so the lining is not lost; the embryo stays implanted and the placenta can form, so there are no periods during pregnancy

In-class work · Part B

- 1 Name the four stages of the cycle.
- 2 Match each hormone to its job.
- 3 Read a table of hormone levels and explain it.

For data: describe the trend first, with numbers.

From intercourse to fertilisation

Teaching line, not a question slide.

Order: B, E, D, A, C.

Q card C is fertilisation; it happens in the oviduct

Say it:

intercourse = IN-ter-kors

ejaculation = ih-jak-yoo-LAY-shun

Put the cards in order.

- A** Sperm swim through the cervix and uterus into the oviducts.
- B** The penis fills with blood and becomes erect.
- C** One sperm enters the egg, and the two nuclei fuse.
- D** Semen is released into the vagina. This is ejaculation.
- E** The penis is put into the vagina.

Which card is fertilisation? Where does it happen?

Zygote, embryo, foetus

Teaching line, not a question slide.

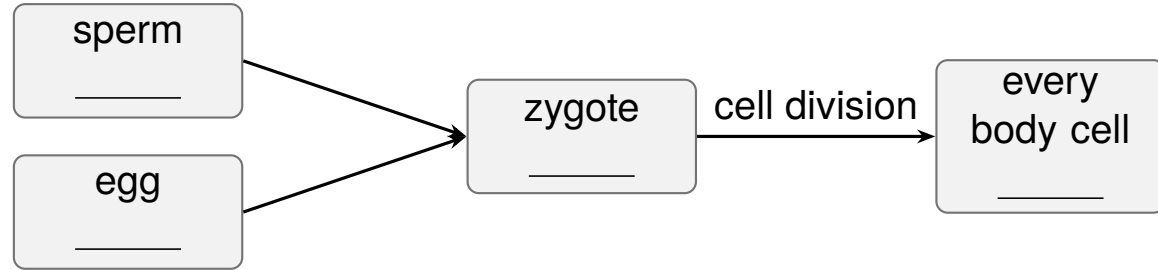
Gaps: zygote; embryo; implantation; foetus.

Q the membrane of the egg changes as soon as one sperm has entered

- 1** The sperm nucleus fuses with the egg nucleus. The new cell is a _____.
- 2** It divides many times as it moves along the oviduct. The ball of cells is an _____.
- 3** The ball of cells sinks into the lining of the uterus. This is _____.
- 4** After about 8 weeks it has all its organs. Now it is a _____.

What stops a second sperm from entering the egg?

Chromosomes: count them



Chromosomes carry the genes. They are in the _____ of every cell.

Why must a sex cell have only half the number?

Teaching line, not a question slide.

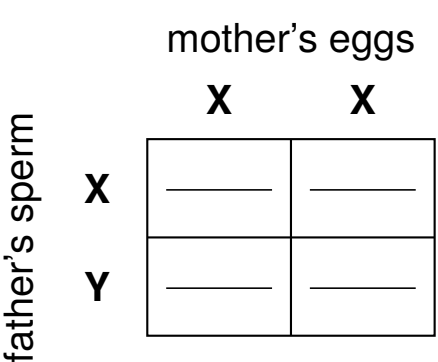
Gaps: sperm 23; egg 23; zygote 46; every body cell 46 (23 pairs); nucleus.

Q so that the zygote has the full 46 and not 92 when the two sex cells join; half comes from each parent

Say it:

genes = JEENZ

Boy or girl?



Female cells have XX. Male cells have XY.

Every egg carries _____. A sperm carries _____ or _____.

- 1 Complete the square.
- 2 What is the chance of a boy?
- 3 Which parent's sex cell decides?

Teaching line, not a question slide.

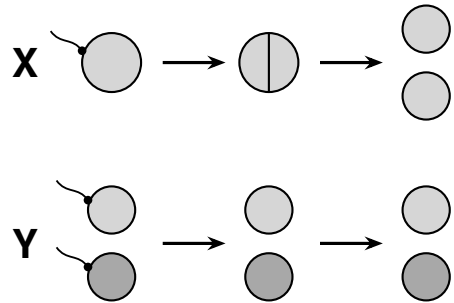
Gaps: X; X or Y.

1. top row XX, XX; bottom row XY, XY

2. 50 % (2 of the 4 boxes)

3. the father's: a sperm with X gives a girl, a sperm with Y gives a boy

Twins



Each row ends with two embryos.

1 Which row shows identical twins?

2 Row X: _____ egg, _____ sperm.

The zygote _____.

3 Row Y: _____ eggs, _____ sperm.

4 Which twins can be a boy and a girl?

Teaching line, not a question slide.

1. row X

2. one, one; splits into two

3. two, two

4. row Y, the non-identical (fraternal) twins: they have different genes; identical twins have the same genes, so they are always the same sex

Say it:

identical = eye-DEN-tih-kul

fraternal = fruh-TUR-nul

Workbook: true or false?

- a** Identical twins happen when a fertilised egg cell divides in two, but the two cells split apart.
- b** Identical twins will always be different sexes.
- c** Non-identical twins happen when two egg cells are fertilised.
- d** Non-identical twins can be the same sex or different sexes.
- e** Identical twins happen when two egg cells are fertilised.
- f** Non-identical twins happen when egg cells turn into embryos on their own.

- (a) true
- (b) false: they are always the same sex
- (c) true

- (d) true
- (e) false: that gives non-identical twins
- (f) false: each egg must be fertilised by a sperm

Workbook: fertilisation

Q1 in the oviduct (fallopian tube)

Q2 they die and are broken down

Q3 on land the sex cells would dry out; inside the body the sperm

can swim in fluid, the sex cells are protected, and a sperm is much more likely to meet the egg

Say it:

evolutionary = ee-vuh-LOO-shun-ree

Q1 Where does fertilisation occur?

Q2 Only one sperm cell joins with the egg cell. What do you think happens to all the other sperms?

Q3 Humans have internal fertilisation as a result of moving onto the land in our evolutionary history. Why do you think this is an advantage on the land?

A boy next time? Write it in this order

Teaching line, not a question slide.

Gaps: X; Y; 50 % (1 in 2); chance.

Each fertilisation is a new event with a new sperm cell.

A couple have three girls. What is the chance that their fourth child is a boy?

- 1 Every egg carries an _____ chromosome.
- 2 Half of the sperm carry X and half carry _____.
- 3 So at every fertilisation the chance of a boy is _____.
- 4 The children born before do not change the _____.

A: the number **M:** lines 1–2 **E:** line 4 as well

Spot the error (3)

Explain where a baby's chromosomes come from, and how twins can be identical.

Sam's answer

A sperm has 46 chromosomes and an egg has 46, so the baby gets all of them. The mother's egg decides if the baby is a boy or a girl. Identical twins form when two eggs are fertilised by two sperm.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. a sperm and an egg have 23 chromosomes each, so the zygote has 46 (L1)
2. the sperm decides the sex: it carries X or Y, and every egg carries X (L7)
3. identical twins come from one fertilised egg that splits; two eggs

give non-identical twins (L11)

Short solution:

A sperm has 23 chromosomes and an egg has 23, so the zygote has 46. The father's sperm decides if the baby is a boy or a girl. Identical twins form when one fertilised egg splits into two.

Practice

- Q1** State where fertilisation happens and name the cell that is formed. **[A]**
- Q2** A boy and a girl are twins. Are they identical twins? Explain. **[M]**
- Q3** Explain why a child has features of both parents but is not identical to either of them. **[E]**

Q1 in the oviduct; a zygote

Q2 no: identical twins come from one zygote, so they have the same chromosomes and are the same sex; a boy and a girl must

come from two eggs and two sperm

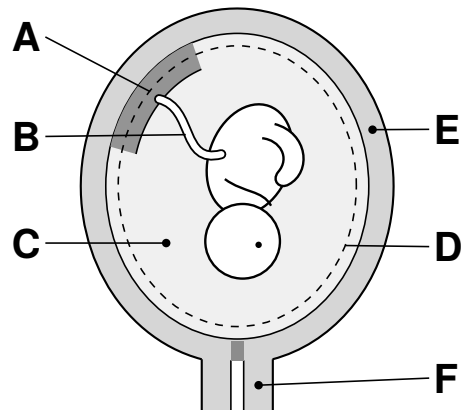
Q3 the child gets 23 chromosomes from the mother's egg and 23 from the father's sperm, so its genes are a mixture of both; the mixture is a new combination, different from each parent

In-class work · Part C

- 1 Put the stages from sex cell to foetus in order.
- 2 Work out the sex of a baby from its chromosomes.
- 3 Explain how two kinds of twins form.

Give the number of chromosomes each time you name a cell.

Quick review: The foetus in the uterus

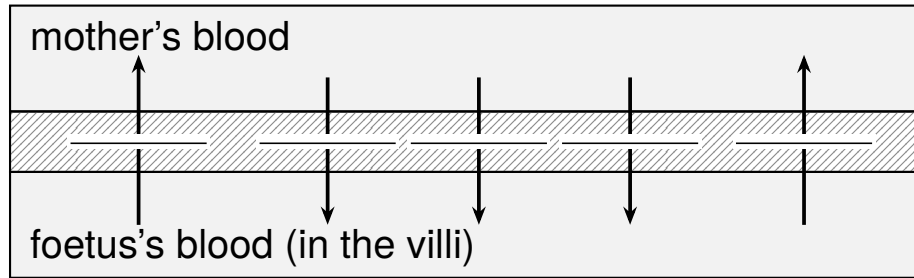


- 1 Name parts A to F.
- 2 Which part cushions the fetus?
- 3 Which part joins the foetus to the placenta?

1. A placenta, B umbilical cord, C amniotic fluid, D amniotic sac, E uterus wall, F cervix
2. the amniotic fluid

3. the umbilical cord
- Say it:
umbilical = um-BIL-ih-kul
amniotic = am-nee-OT-ik

The placenta: what crosses?



Down: three things the fetus needs. Up: two waste products.

The two bloods never _____. Substances cross by _____.

Teaching line, not a question slide.

Down: oxygen; glucose (food, nutrients); antibodies. Up: carbon dioxide; urea.

Gaps: mix; diffusion.

Say it:

villi = VIL-eye

glucose = GLOO-kohz

antibodies = AN-tee-bod-eez

urea = yoo-REE-uh

diffusion = dih-FYOO-zhun

Workbook: the role of the placenta

The placenta develops from foetal tissues and allows the _____ of materials between mother and foetus. It has a large number of blood vessels that do not touch. This ensures there is no _____ of the maternal and foetal blood.

- a** Give three reasons why it is important that the maternal and foetal blood do not mix.
- b** Explain why the embryo needs oxygen and glucose.
- c** Which substances diffuse from the foetus' blood into the mother's blood and why is it important that they are removed?

- (a) the mother and the foetus can have different blood types; the mother's blood pressure is higher and would damage the foetus's small blood vessels; keeping the bloods apart stops many bacteria and harmful substances reaching the foetus
- (b) for respiration, which releases the energy its cells need to divide and grow
- (c) carbon dioxide and urea; they are waste products and would

poison the foetus if they built up
Short solution:
Blanks: exchange (diffusion); mixing.
Say it:
foetal = FEE-tul
maternal = muh-TUR-nul
respiration = res-pih-RAY-shun

Oxygen to the foetus: write it in this order

- 1 The mother's blood has a _____ concentration of oxygen than the foetus's blood.
- 2 Oxygen _____ from high to low concentration, across the thin wall.
- 3 The villi give a large _____, so diffusion is fast.
- 4 The foetus's blood carries the oxygen away in the umbilical _____.

A: line 2 **M:** lines 1–2 **E:** lines 3–4 as well

Now tell the same story for carbon dioxide.

Teaching line, not a question slide.

Gaps: higher; diffuses; surface area; vein.

Q the foetus's blood, arriving in the umbilical artery, has a higher concentration of carbon dioxide than the mother's blood, so carbon

dioxide diffuses across the villi into the mother's blood, which carries it away

Say it:

concentration = kon-sun-TRAY-shun

artery = AR-tuh-ree

Harm: what else crosses the placenta?

From the mother Effect on the foetus

cigarette smoke	less _____ reaches it; the baby is smaller
alcohol	damages the developing _____
drugs	the baby can be born addicted
rubella virus	deafness; heart and eye defects

Why can even a small amount do a lot of damage?

Teaching line, not a question slide.

Gaps: oxygen; brain.

Q the foetus is very small and is growing very quickly

Cigarette smoke: carbon monoxide takes the place of oxygen in the mother's blood; nicotine also crosses the placenta.

Say it:

rubella = roo-BEL-uh

addicted = uh-DIK-tid

monoxide = mon-OK-syd

nicotine = NIK-uh-teen

Pregnancy in numbers

Teaching line, not a question slide.

Gaps: 40; 8 (the glossary says after 8 weeks, page 28 says nine);
20; 38.

Q premature

After 20 weeks the loss of a baby is called a stillbirth.

Say it:

gestation = jes-TAY-shun

miscarriage = MIS-ka-rij

premature = PREM-uh-chur

- 1 Pregnancy (gestation) lasts about _____ weeks, counted from the first day of the last period.
- 2 The embryo is called a foetus from about week _____.
- 3 Losing a foetus before _____ weeks is a miscarriage.
- 4 A baby born before _____ weeks is premature.

A baby is born at 35 weeks. Which word describes this birth?

Birth: three stages

Teaching line, not a question slide.

Order: B, C, A. Stage 1 labour; stage 2 delivery; stage 3 afterbirth.

Q the amniotic fluid

Say it:

dilates = dy-LAYTS

labour = LAY-bur

Put the cards in order. Then name each stage.

A The placenta is pushed out.

B Muscles of the uterus contract. The cervix opens (dilates). The waters break.

C The baby is pushed out through the vagina. The cord is tied and cut.

Stage 1 _____ Stage 2 _____ Stage 3 _____

What are “the waters”?

Workbook: fertility problems and solutions

Problems	Solutions
Low sperm count	Injecting sperm cell into egg cell, then IVF
Oviducts too narrow	
Sperm ducts blocked	
Lining stops implantation	
Few ripe eggs released	
Sperm do not swim well	

- 1. doctors try to widen the oviducts; or IVF
- 2. doctors try to widen the sperm ducts
- 3. IVF
- 4. IVF

5. inject a sperm cell into an egg cell with a fine needle
Short solution:
These are the answers in the paragraphs on page 31. The numbers 1 to 5 are the five empty rows, from the top.

IVF: fertilisation in a dish

Put the cards in order.

- A** Eggs are collected from the woman's ovaries.
- B** An embryo is put into the uterus, where it may implant.
- C** The woman is given hormones, so several eggs mature.
- D** The eggs are mixed with sperm in a dish. Some are fertilised.
- E** The embryos grow for about two days.

Why are twins more common after IVF?

Teaching line, not a question slide.

Order: C, A, D, E, B.

Q often more than one embryo is put into the uterus, and each one

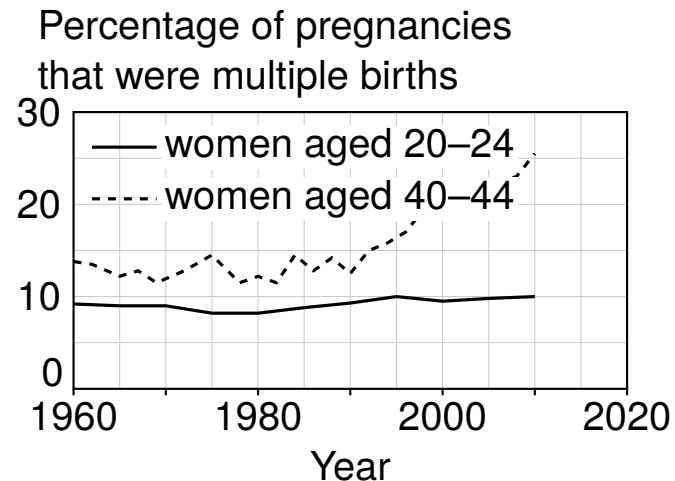
can implant

In vitro means in glass: fertilisation happens outside the body.

Say it:

vitro = VEE-troh

Workbook: question 3, describe



- (a) Describe what has happened to the number of multiple births in women aged 40-44.
- (c) Describe what happened to the number of multiple births in women aged 20-24.

Q3 (a) from 1960 to about 1990 it stayed between about 11 % and 15 %; after 1990 it rose steadily to about 25 % in 2010
Q3 (c) it stayed almost the same, at about 9 % to 10 %, from 1960 to

2010

Short solution:

A multiple birth is two or more babies born from one pregnancy.

Workbook: suggest and predict

Headlines: “Fertility treatment shown to increase chances of multiple birth” · “Older women more likely to have problems becoming pregnant” · “Fertility treatment becoming more common”

Q3 (b) Suggest a hypothesis for why this has been happening.

Q3 (d) Suggest a hypothesis for why this pattern is not like that for women aged 40–44.

Q4 (a) Predict the percentage of multiple births last year for women aged 40–44.

Q3 (b) fertility treatment has become more common; older women are more likely to need it; fertility treatment increases the chance of a multiple birth

Q3 (d) women aged 20 to 24 rarely have problems becoming pregnant, so few of them have fertility treatment

Q4 (a) about 30 % or a little more: any value above 25 % that continues the rising line

Q4 (b) I extended the trend line from 1990 to 2010 forward to last year

Say it:

hypothesis = hy-POTH-uh-sis

Spot the error (4)

Describe how a foetus gets oxygen and how it is protected in the uterus.

Sam's answer

The mother's blood flows through the umbilical cord into the foetus. Oxygen moves by diffusion from a low concentration to a high concentration. The amniotic fluid gives the foetus its food and protects it from bumps.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. the two bloods never mix: only the foetus's blood flows in the umbilical cord (L9)
2. diffusion is from a high concentration to a low concentration (L10)
3. the amniotic fluid does not feed the foetus: food comes from the mother's blood through the placenta (L13)

Short solution:

Oxygen diffuses from the mother's blood into the foetus's blood in the placenta, from a high concentration to a low concentration. The foetus's blood carries it along the umbilical cord. The amniotic fluid protects the foetus from bumps.

Practice

Q1 Name two substances that pass from the foetus's blood to the mother's blood. **[A]**

Q2 Explain why a pregnant woman who smokes is likely to have a smaller baby. **[M]**

Q3 IVF clinics now usually put one embryo into the uterus, not three. Suggest why, and explain how this changes the number of twins and triplets. **[E]**

Q1 carbon dioxide and urea

Q2 carbon monoxide from the smoke is carried in the mother's blood, so less oxygen crosses the placenta; the foetus's cells

respire less and release less energy, so it grows more slowly

Q3 a multiple birth is riskier for the mother and the babies, who are often premature and small; with one embryo only one can implant, so there are fewer twins and triplets

In-class work · Part D

- 1 Label the foetus in the uterus.
- 2 Explain how a waste product leaves the foetus.
- 3 Describe and explain a table about the foetus.

Name the substance, the direction and the process.

Ways to lose marks

Say the correct version of each one.

- 1 “A sperm cell has 46 chromosomes.”
- 2 “Fertilisation happens in the uterus.”
- 3 “Ovulation is when the lining is lost.”
- 4 “The mother’s blood flows into the foetus.”
- 5 “A sperm has a tail.” (asked to explain an adaptation)
- 6 “It went up because of IVF.” (asked to describe a graph)

Teaching line, not a question slide.

1. 23: every sex cell has one set; the zygote and body cells have 46 (L1)
2. in the oviduct; the embryo implants in the uterus later (L2)
3. ovulation is the release of an egg; losing the lining is

menstruation (L5)

4. the bloods never mix; oxygen and food diffuse across the placenta (L9)
5. add how it helps: so that it can swim to the egg (L3)
6. describe needs the trend and numbers from the graph; the reason is for suggest or explain (L12)

Summary

Teaching line, not a question slide.

Ask her to say each line with the book closed, then give one example question for each.

- 1** Sex cells have 23 chromosomes; fertilisation in the oviduct makes 46.
- 2** Cycle: period, lining builds up, ovulation on day 14, lining kept thick.
- 3** Placenta: diffusion from high to low; the two bloods never mix.
- 4** Explain = fact + because. Describe a graph = trend + numbers.

In-class work · Part E

12 minutes of writing, no notes, no booklet.

Then we mark it together: A, M or E for each question.

Read the command word first: state, describe, explain, suggest.

Homework: two parts, about 40 minutes each

- 1 Part 1: sex cells, the two systems, the menstrual cycle.
- 2 Part 2: fertilisation, chromosomes, pregnancy, fertility.

Do the two parts on two different days.